RUDRA SHARMA

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SUMMARY

Backend engineer with around a year of software development experience across two internships, building Java/Spring Boot and Node.js microservices. Hands-on with event-driven systems (Kafka, BullMQ), multi-tenant SaaS, Redis caching, Keycloak/JWT auth, and AWS (EC2, S3).

EXPERIENCE

EsMagico **June 2026 – Present**
Software Engineer Intern *Remote, India*

- Own **3 Spring Boot microservices** in a License & Certificate Management System built on eGov's DIGIT platform: Verifiable Credentials, PDF generation, and JSON Schema registry.
- Built **Kafka producers and consumers** for event-driven inter-service communication; the certificate service consumes **payment-completed events** to drive application workflow state transitions.
- Designed the Verifiable Credentials service with **2 pluggable credential adapters** (**walt.id**, **MOSIP**), enabling standards-based credential issuance and verification.
- Built a PDF service that stores JSON templates, renders them to HTML, and generates PDFs via **headless Chromium at $\sim$5 PDFs/sec**.
- Built a schema registry service for managing the JSON Schemas used across services.
- Containerized services with **Docker**; supported DevOps on **Kubernetes** deployments, shipping through **Jenkins CI/CD** pipelines.

Tryidol Tech **Sep 2025 – Feb 2026**
Software Engineer Intern *Bhopal, India*

- Built backend services (Node.js, Express, MySQL) for a **multi-tenant SaaS ERP** serving **45 tenants** across **5 modules**, including Attendance (AMS), HRMS, User Management (UMS), and PMS.
- Implemented event-driven async workflows with **BullMQ** job queues and Node.js **EventEmitter** for event-based notifications, keeping heavy work off the request path.
- Built multi-tenant file storage on **AWS S3** with **presigned-URL uploads** and per-tenant bucket management.
- Cached analytics dashboards in **Redis**, cutting response time on million-row queries from **10s+ to 1–2s**.
- Designed REST APIs secured with **JWT and RBAC** on a normalized relational schema that kept data consistent across modules.
- Built async XML ingestion for a job aggregation platform processing **10M+ listings in under 15s** per run, using a native C++ parser across Node.js **worker threads**.
- Built job filtering and short-link tracking to capture user interaction data for analytics and fraud detection.

PROJECTS

DoubtNix | **AI-Powered Doubt Solving Platform [live]** *Node.js, Gemini API, AWS*

- Built and deployed a full-stack document Q&A platform returning LLM-generated answers via the **Gemini API**; backend hosted on **AWS EC2**.
- Moved document parsing and post-processing into an **async event-driven pipeline**, decoupling heavy work from the request cycle.
- Secured with **3-layer auth** (JWT + OTP + 2FA) and session management; Cloudinary-based file storage; React + TanStack Query frontend.

Sync-Chat | **Real-Time Chat Application [live]** *Node.js, Socket.IO, Redis, AWS*

- Built 1:1 and group messaging with **Socket.IO** and a **Redis Pub/Sub adapter** so events sync across multiple server instances for horizontal scaling; backend on **AWS EC2**.
- Implemented JWT cookie auth with **refresh-token rotation** and **3-role** group RBAC (owner/admin/member).
- Built multi-device presence tracking, typing indicators, and **VAPID web push** notifications for offline users.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, SQL
Backend: Spring Boot, Node.js, Express.js, REST APIs, Microservices, Event-Driven Architecture
Messaging & Async: Kafka, BullMQ, Socket.IO
Data & Caching: PostgreSQL, MySQL, MongoDB, Redis
Cloud & DevOps: AWS (EC2, S3), Docker
Security: Keycloak, JWT, RBAC, Rate Limiting, OTP/2FA
Frontend & Tools: React.js, TanStack Query, Git, Postman

EDUCATION

IES University – B.Tech in Computer Science and Engineering 2022 – 2026

ACHIEVEMENTS & CERTIFICATIONS

- Solved **300+ DSA problems** across LeetCode, GFG, and Coding Ninjas; **6+ open-source contributions** during Hacktoberfest.
- Certifications: Object-Oriented Programming (Coding Ninjas); Postman API Fundamentals Student Expert.